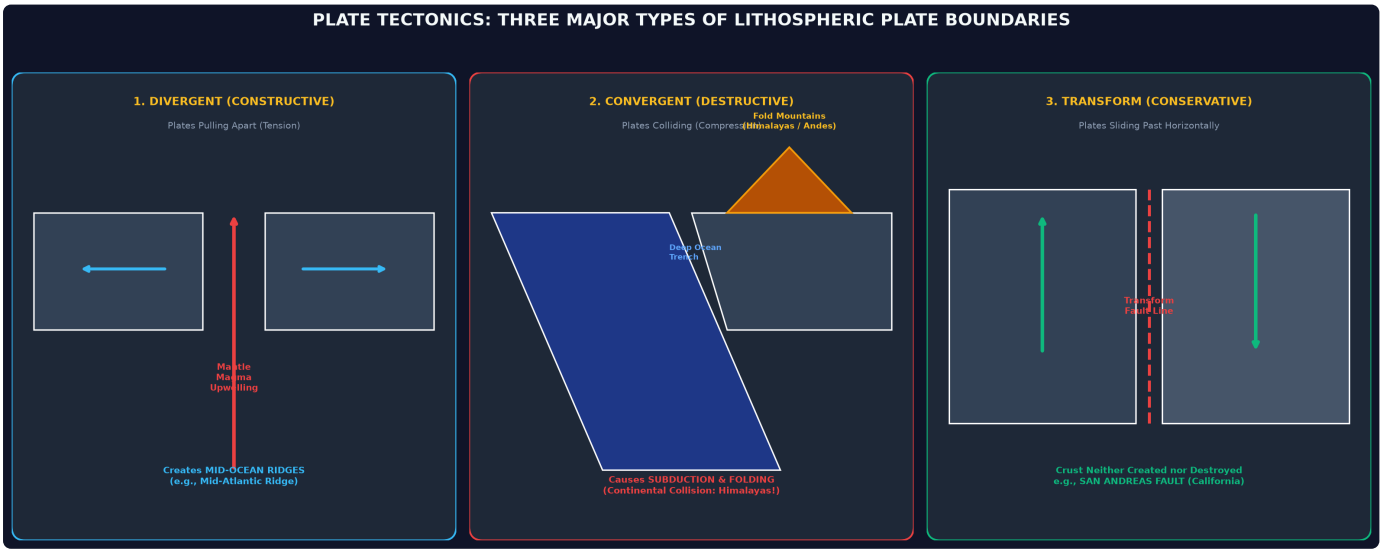
Topic 15 — Geomorphology & Landform Processes

Sources baked in: NCERT Class 11 (Geomorphic Processes, Landforms), PWOnlyIAS / Drishti geomorphology notes, Lucent landform match-lists, **Ghatnachakra / multi-PCS Extra** (weathering · fluvial · glacial · aeolian · coastal · karst · drainage · deltas · **CB–73+ valleys** · **CB–143+ river landforms**), **UKPCS** Prelims stems (rocks · landforms), UPPCS Prelims 2018–2025 **Weight:** ★★★ — trellis, bird's-foot (Mississippi only) vs arcuate, moribund Bengal vs Cauvery, Imphal lacustrine, Damodar fault-valley, rock-cycle pairs; teach full process map for **UPPCS + UKPCS Last verified:** September 2026 — Extra Drill expanded with CB–73+ / CB–143+ (valleys · fluvial landforms) + UKPCS rock stems **Current Affairs:** none mandatory (process chapter); Himalayan fossils / delta types stay static

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Static	Trellis / bird's-foot / Imphal lacustrine surface	Process stems stay map-static	NCERT / coaching
Static	Mekong delta = southern Vietnam (not Cambodia)	Country trap in delta stems	Atlas
Static	Moribund wording: Bengal subdivision vs Cauvery lobe	Two different stem shapes	Coaching
Static	Himalaya fossil belts (Himadri / Lesser / Shiwalik)	Mixed landform–fossil stems	NCERT

N.0 Framework — Geomorphic Processes



Geomorphology studies how landforms are made and destroyed.

- **Endogenic processes** come from Earth's interior — folding, faulting, volcanism, earthquakes and uplift. They build relief.
- **Exogenic processes** work at the surface — weathering, mass wasting, erosion, transport and deposition by river, wind, glacier, sea and groundwater. They wear relief down and fill basins.
- **Gradation** is the overall smoothing of the land: **degradation** (wearing down) plus **aggradation** (building up by deposition).
- Davis framed a **cycle of erosion** as structure–process–time (youth–mature–old). Penck and King modify the timing, but the landform vocabulary stays useful.
- **Structure** is the rock arrangement (folds, faults, joints, bedding). **Process** is the work of agents. **Time** is how long the process has operated on that structure.
- Weathering prepares loose material **in place**.
- Erosion and transport move it.
- Deposition rebuilds new landforms downstream or downslope.
- Different climates favour different agents.
- Humid belts emphasise running water.
- Arid belts emphasise wind.
- Cold highlands emphasise ice.
- Coasts emphasise waves.
- Limestone belts emphasise groundwater solution.

Order of this chapter

1. Rocks and the rock cycle supply the raw (रॉ) material.
2. Weathering prepares material; erosion and deposition move and drop it.
3. Each agent (fluvial, glacial, aeolian, coastal, karst) leaves a signature landform set.
4. Structure (folds, faults, basins) steers drainage patterns and valley types.

Teach the **full process map** — next year can ask any standard NCERT agent or pair, not only trellis or Damodar.

N.1 Rocks — Igneous, Sedimentary, Metamorphic

Rocks are the solid building material of the crust. Notes ask **class**, **parent–daughter pairs**, and **intrusive body shapes**.

Class	How formed	rocks
Igneous (primary)	Cooling of magma/lava	**Granite** (intrusive, acidic, coarse); **basalt** (extrusive, basic, Deccan Trap (दक्कन ट्रैप)); gabbro, pegmatite, pumice, obsidian
Sedimentary (secondary)	Compaction of debris / chemical / organic	Sandstone, shale, **limestone**, conglomerate, coal, loess beds — **fossils possible**
Metamorphic	Heat + pressure on existing rock	See pair table

Metamorphic pairs (mug)

Parent Rock	Metamorphic Product	Type / Notes
Limestone / dolomite	**Marble**	Sedimentary → Metamorphic (calcite recrystallised)
Sandstone	**Quartzite**	Sedimentary → Metamorphic (quartz grains welded)
Granite	**Gneiss**	Igneous → Metamorphic (banded crystalline)
Gabbro	**Serpentine**	Igneous → Metamorphic
Basalt	**Hornblende / Amphibolite / Schist**	Igneous → Metamorphic
Shale / Clay	**Slate → Phyllite → Schist**	Progressive foliation series
Coal	**Graphite / Diamond**	Organic Sedimentary → Metamorphic

Inline PYQ — UPPCS Prelims 2019, Q11

With reference to the Himalaya, which of the following statements is/are correct?

1. The Greater Himalaya / Himadri is largely crystalline and fossil-poor.
2. The Lesser Himalaya contains marine fossils in sedimentary rocks.
3. The Outer Himalaya / Shiwalik contains fossils including human remains.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Show answer

Ans: D

Igneous — acidic / basic & texture

	Acidic (SiO_2 high)	Basic (SiO_2 low)
Intrusive (plutonic, coarse)	**Granite**	Gabbro
Extrusive (volcanic, fine)	Rhyolite	**Basalt** (Deccan Trap)
Glassy / frothy	Obsidian	Pumice (floats)

Porphyritic texture shows large crystals in a fine groundmass from two-stage cooling. **Pegmatite** forms very coarse veins. **Intrusive bodies (NCERT match)**

Body	Role
Batholith	Huge granite mass; floor not seen; mountain cores
Laccolith	Lens, **convex up**; arches overlying beds
Lopolith	Saucer, **concave up**
Phacolith	Lens in fold crest/trough
Sill	Sheet **parallel** to bedding
Dyke	Wall that **cuts** bedding (discordant (असंगत))
Volcanic neck / plug	Filled pipe left standing

- A fissure flood-basalt plateau is the **Deccan Trap**.
- Intertrappean beds between lava flows are **freshwater**, not marine fossils.

Inline PYQ — UPPCS Prelims 2024, Q59

Intertrappean beds of the Deccan Trap are associated with:

A. Marine fossils only

B. Freshwater conditions / fossils

C. Only glacial till

D. Only aeolian dunes

Show answer

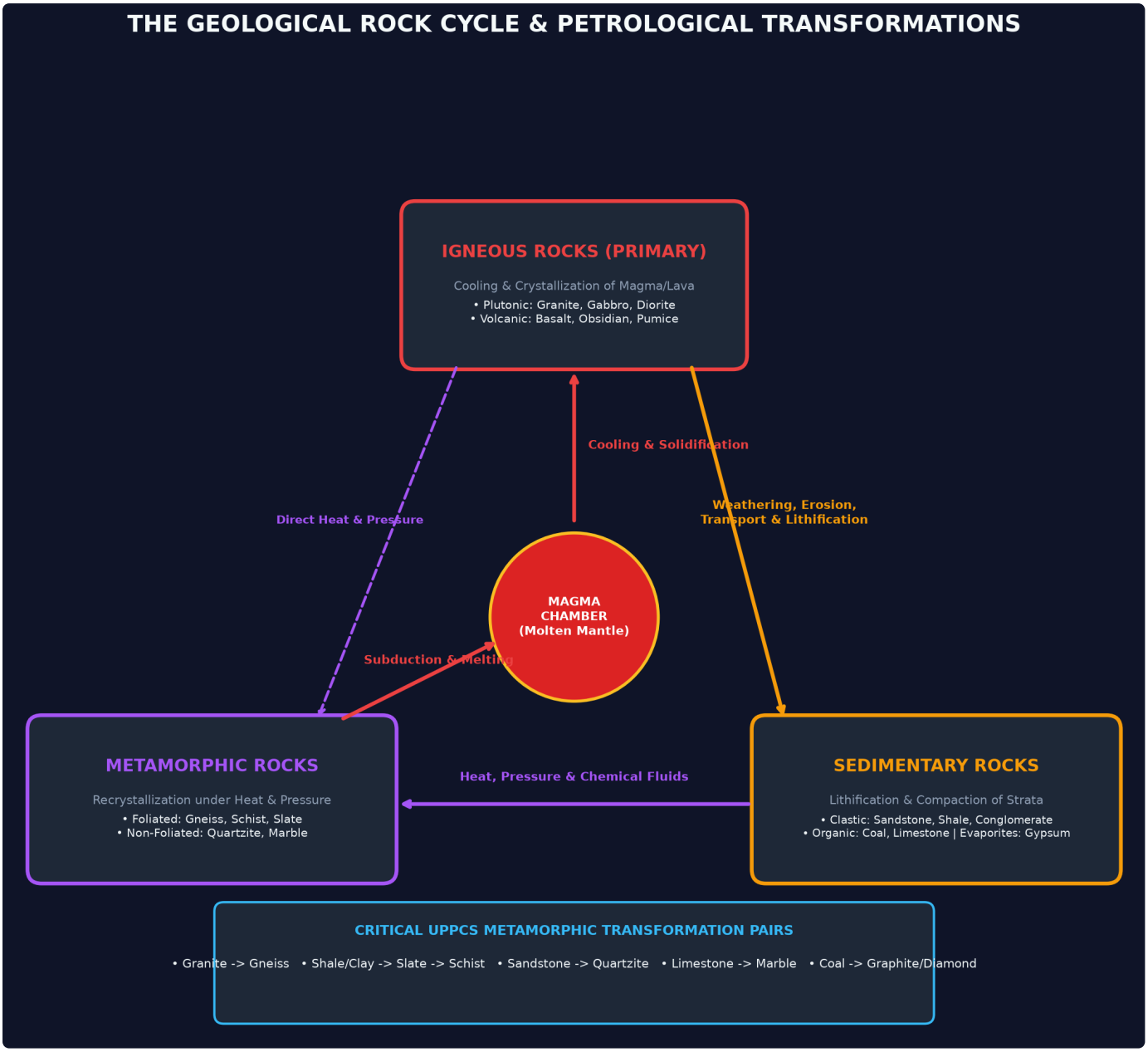
Ans: B

Sedimentary families

Family	Rocks
Clastic	Conglomerate (**rounded**), **breccia** (angular), sandstone, shale/mudstone
Chemical	Limestone, dolomite, gypsum, rock salt (लवणाध्यक्ष), chert
Organic / biogenic	Coal, peat, chalk, some limestone

Fossils appear almost only in sedimentary rocks. Loess beds are wind silt later lithified. **Foliated** metamorphic fabric climbs from slate to schist to gneiss (banding). **Non-foliated** metamorphic rocks include marble and quartzite.

N.2 Rock Cycle



The **rock cycle** is the continuous pathway that turns magma and rock from one class into another.

- **Magma** cools and solidifies to form **igneous** rock — intrusive granite deep underground, or extrusive basalt at the surface.
- Weathering breaks that rock in place.
- Erosion carries the debris.
- Deposition and lithification build **sedimentary** rock such as sandstone, shale or limestone.
- Heat and pressure without full melting turn igneous or sedimentary rock into **metamorphic** rock.
- Limestone becomes marble.
- Sandstone becomes quartzite.
- Shale climbs the slate–schist–gneiss ladder.
- If temperature rises enough, any rock can **melt** again into magma and restart the cycle.

- The cycle is not one-way: igneous rock can metamorphose without becoming sediment first, and sedimentary rock can melt without a long metamorphic stay.
- Uplift brings deep rocks to the surface so weathering can attack them again.
- Burial takes surface sediment down into zones of pressure and heat.

From	Process	To
Magma	Cooling / crystallisation	Igneous
Any exposed rock	Weathering → erosion → deposition → lithification	Sedimentary
Igneous or sedimentary	Heat + pressure (no full melt)	Metamorphic
Any rock	Melting	Magma

> **Logic:** ** Weathering alone does **not** make sedimentary rock. Sediment must also be deposited and lithified.

N.3 Weathering, Erosion, Deposition

These four (चातुर्यम) words are traps if swapped.

- **Weathering** breaks rock **in place** — no travel.
- **Erosion** detaches and **transports** material by an agent.
- **Deposition** drops the load when energy falls.
- **Denudation** is weathering plus erosion lowering the land.
- **Mass wasting** is gravity movement (creep, slide, fall) without a true transporting stream.

Term	Meaning
Weathering	In situ breakdown
Erosion	Pickup + transport
Deposition	Drop of load
Denudation	Weathering + erosion
Mass wasting	Gravity movement

Physical / mechanical weathering

Physical weathering cracks and peels rock without changing its chemistry.

- **Frost / ice wedging** works where water freezes in cracks: ice expands and pries the rock apart (cold, wet climates).
- **Exfoliation / unloading** peels rock in **onion-like sheets** when overlying pressure is removed — classic on **granite** domes.

- **Thermal / insolation (सूर्यताप)** weathering follows desert day–night heating and cooling; the surface expands and contracts until it flakes.
- **Salt wedging** grows salt crystals in pores on arid coasts and playas and pries grains apart.
- **Block disintegration** breaks jointed masses into large angular blocks; **granular disintegration** frees mineral grains one by one (common in coarse granite).
- **Spheroidal weathering** rounds corestones as chemical softening and physical peeling work together from the outside in.

Process	Quick recall
Frost / ice wedging	Water freezes and pries cracks
Exfoliation / unloading	Onion sheets on granite
Thermal / insolation	Desert day–night flake
Salt wedging	Crystal growth in arid pores
Block vs granular	Joint slabs vs grain-by-grain
Spheroidal	Rounded corestones

Chemical weathering (fastest where it is **hot and wet**)

Chemical weathering changes minerals; humid tropics are the classic speed zone.

- **Carbonation** forms when CO_2 in rain makes weak carbonic acid that dissolves **limestone** and builds **karst**.
- **Solution** dissolves highly soluble rocks such as rock salt and gypsum directly into water.
- **Oxidation** rusts iron-bearing minerals and often colours soils red or brown.
- **Hydration** lets minerals take up water, swell, and crack the surrounding rock.
- **Hydrolysis** breaks feldspar into **clay** minerals — a main route from granite to clay soils.
- **Chelation** uses organic acids from plants and microbes to pull metal ions out of minerals (biological overlap).

Process	Quick recall
Carbonation	Carbonic acid → limestone karst
Solution	Direct dissolve (salt, gypsum)
Oxidation	Iron rusts; red soils
Hydration	Water uptake → swell and crack
Hydrolysis	Feldspar → clay
Chelation	Organic acids leach metals

Biological weathering adds roots that wedge joints, burrows that open rock, and lichen or moss acids that etch surfaces.

Residual weathered mantle becomes soils. **Laterite** (लेटराइट) forms by intense leaching in the tropics. Running water progresses from sheet to rill to gully and, in the classic case, to a **Chambal ravine**. **Mass wasting** (gravity) includes fall, slide, slump, flow, **creep**, solifluction (thaw), mudflow and debris flow.

N.4 Agents of Erosion

Exogenic agents sculpt the land by picking up, carrying and dropping rock debris. Each agent has a work method and a signature landform set.

Agent	Work words	Signature
River	Hydraulic, abrasion/corrasion, attrition, solution; load = traction · saltation · suspension · solution	V-valley → meander → delta
Wind	Deflation, abrasion, attrition	Barchan, yardang, loess
Glacier	Plucking, abrasion	U-valley, cirque, moraine
Sea	Hydraulic, abrasion, corrosion, attrition	Cliff, stack, spit
Groundwater	Solution	**Karst** in limestone

Running water (fluvial)

- Rivers cut steep **V-shaped valleys** in youth by hydraulic action and abrasion.
- Load moves as traction, saltation, suspension and solution as the channel carries bed and suspended material.
- As gradient eases, the river widens into meanders, floodplains and natural levees.
- At a lake or sea mouth, slowing water builds a **delta** of layered alluvium.

Wind (aeolian)

- Wind works best in dry, sparsely vegetated lands.
- **Deflation** lifts fine dust and lowers desert floors.
- **Abrasion** sandblasts rock into polished surfaces and elongated **yardangs**.
- Deposition builds crescent **barchan** dunes where sand supply and wind direction are steady.
- Thick **loess** blankets form where wind drops fine silt far from the source.

Glacier

- Moving ice **plucks** jointed blocks and **abrades** the floor into striations and rock flour.
- High mountain erosion leaves **cirques**, arêtes and horns above glacial troughs.
- The main trough itself is a broad **U-shaped valley**.
- Melting ice drops the load as lateral, medial and terminal **moraines**, with outwash beyond the snout.

Waves (coastal / marine)

- Waves attack cliffs by hydraulic smash, abrasion and attrition.

- Retreat of a cliff leaves a wave-cut platform, caves, arches and isolated **stacks**.
- Longshore drift builds beaches, **spits** and bars where shallow water favours deposition.
- Soft coasts retreat faster; hard headlands persist and shape the shoreline plan.

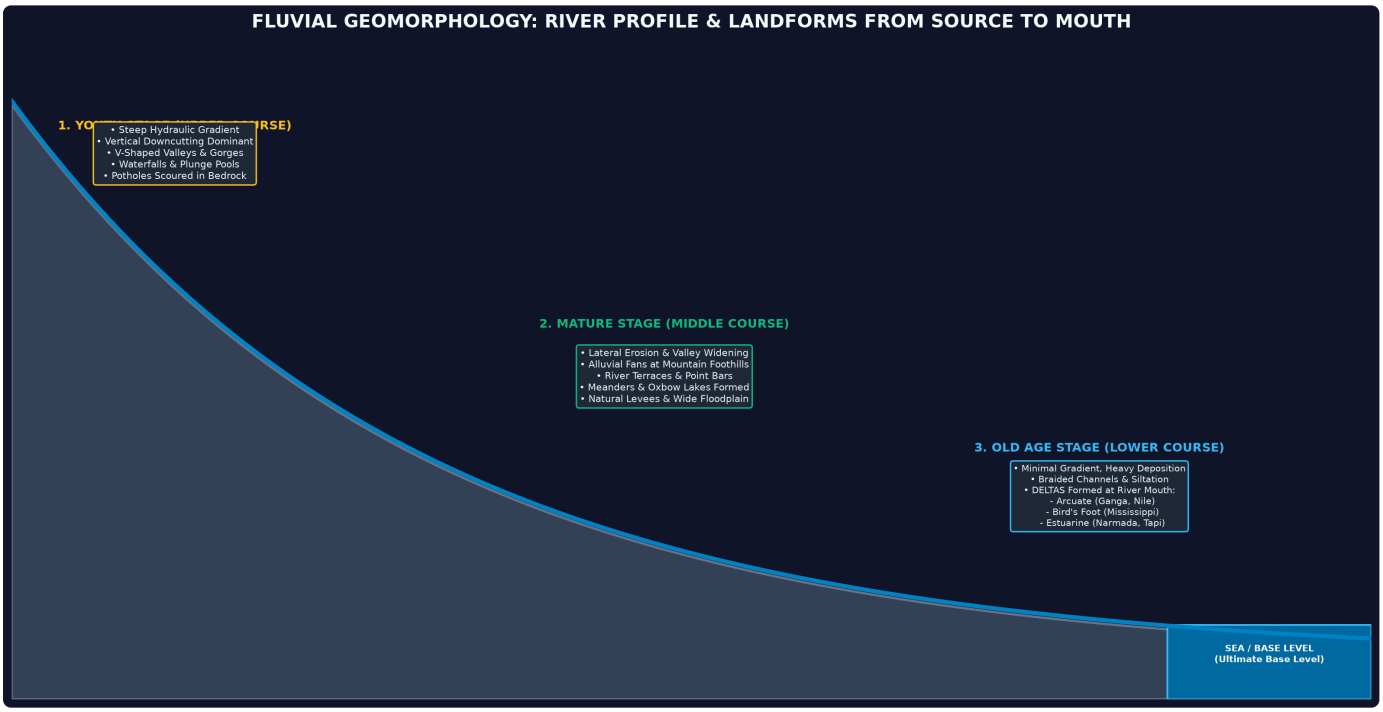
Groundwater

- Underground solution is strongest in soluble **limestone** and dolomite.
- Sinkholes, disappearing streams and caves form the classic **karst** suite.
- Stalactites grow from cave roofs and stalagmites rise from floors where dripstone is deposited.
- Roof collapse can widen closed depressions into larger karst lowlands.

Agents often work together. A river may inherit a glacial trough, and wind may rework beach sand into dunes.

> **Logic:** ** Weathering prepares rock in place; an **agent of erosion** must also transport. Do not treat frost wedging alone as “river work.”

N.5 Fluvial Landforms (Davis cycle)



Rivers cut, carry and deposit. Davis stages are a teaching ladder, not a strict clock for every valley.

Stage	Landforms
Youth	Steep **V-valley**, gorge/canyon, waterfall, rapids, **potholes**, interlocking spurs
Mature	**Meander**, slip-off slope + river cliff, **floodplain**, **natural levee**, bluff, widening valley
Old	Broad floodplain, **ox-bow** / mortlake / meander scar, yazoo, **delta**, deferred tributary
Rejuvenation	Uplift / base-level fall → **knickpoint**, river **terrace**, **incised meander**

Fluvial forms — Teaching Card

Meanders and floodplain suite

- A **meander** is a river loop on a gentler gradient.
- On the **concave** outer bank the current cuts a **river cliff**; on the **convex** inner bank it drops a **point bar** (slip-off slope).
- A **floodplain** grows as point bars accrete and floods drop overbank silt beyond the channel.
- A **natural levee** is the raised ribbon of coarser silt left along the channel edge after floods.
- An **ox-bow** (mortlake) forms when a meander neck is cut off and the abandoned loop becomes a lake — common on the **Ganga** (गंगा) plain; a dry scar marks an older cut-off.
- A **yazoo** tributary is **deferred** behind the main levee: it runs parallel to the master stream and joins only farther downstream.

Fans, braids, and youth cuts

- **Potholes**: Cylindrical holes drilled into rocky river beds by pebbles whirling in eddies (abrasion/cavitation). Famous Indian pothole fields are seen in Maharashtra in the beds of the **Kukadi river** (Nighoj potholes), **Krishna**, and **Godavari**.
- **Etymology of Delta**: The geographical term **'Delta'** was first coined by Greek historian **Herodotus** (the 'Father of History') after the Greek triangular letter delta (Δ), observing the fan shape at the mouth of the river **Nile**.
- An **alluvial fan** (or cone) is a dump of coarse load at a **mountain foot** where the stream leaves a steep gorge onto a plain.
- When neighbouring fans coalesce along a mountain front they form a **bajada**; the enclosed arid basin floor is a **bolson**.
- A **braided** channel splits into many shallow threads that reunite — the **Brahmaputra** (ब्रह्मपुत्र) is the classic Indian type.
- A **pothole** is drilled in the bed by pebbles spinning in an eddy — a youth-stage signature.
- A **waterfall** can sit on a hard cap rock, a fault scarp, a hanging valley lip, or a **knickpoint** after rejuvenation.

Gorge, canyon, and graded profile

- A **gorge** is a steep-sided mountain cut, often narrow and deep.
- A **canyon** is typically wider, with stair-step walls, and is common in **arid** plateaus (Grand Canyon type).
- A **graded profile** is the smooth concave long-profile a river approaches toward **base level** (usually the sea).

Form	Quick recall
Meander	Outer cut cliff; inner point bar
Ox-bow	Cut-off meander lake
Floodplain	Point-bar + overbank silt
Levee	Raised silt ribbon along channel

Form	Quick recall
Yazoo	Deferred tributary behind levee
Fan / bajada	Mountain-foot dump; coalescing fans
Braided	Many threads (Brahmaputra type)
Gorge vs canyon (गॉर्ज)	Narrow steep cut vs wider arid stair-step

Cycle theories

- **Davis** framed erosion as **structure–process–time**: youth → mature → old age, ending in a humid **peneplain** with leftover **monadnocks**.
- **Penck** argued that uplift and denudation often work **together**, so landscapes need not wait for a neat old-age plain.
- **L.C. King** stressed pediment retreat in arid and savanna (सवाना) lands, ending in a **pediplain** rather than a Davis peneplain.
- Do not mix the three end-products. **Davis** = humid **peneplain**. **King** = arid **pediplain**. **Penck** is about **simultaneous uplift and erosion**, not a third plain name and not King's pediment story.

Theory	Quick recall
Davis	Structure–process–time → **peneplain** + **monadnock**
Penck	Uplift and denudation **together** (no neat old-age wait)
L.C. King	Pediment retreat → **pediplain**

Stream origin names — Teaching Card

Genetic stream names describe **what the river follows**, not how big it is.

- A **consequent** stream follows the **initial slope** of the land after uplift or emergence.
- A **subsequent** stream grows later along **weaker beds** or **faults**; it often cuts strike valleys and helps build a **trellis** pattern.
- An **obsequent** stream flows **opposite** to the consequent — typically down a scarp face.
- A **resequent** stream later flows in the **same direction** as the original consequent, after the landscape has been reworked.
- An **insequent** stream shows **no obvious structural control** — common where rock is uniform and drainage is **dendritic**.
- **Antecedent** drainage is **older than the uplift** it cuts — Himalayan transverse giants (Indus (सिंधु), Sutlej (सतलुज), Brahmaputra and often Ganga in keys) are the classic Indian centre.
- **Antecedent** is genetic history (इतिहास). **Dendritic** is plan-view pattern on uniform rock or alluvium. The Ganga can be **antecedent** in uplift-origin keys and still show **dendritic** drainage on the Ganga–Assam plain — different questions, not a contradiction.
- **Superimposed** drainage is **inherited from a cover** that is now stripped — the **Chambal** (चंबल) is the classic Indian example.
- **Concordant** (संगत) drainage follows structure and slope; **discordant** drainage cuts across structure.

- In **river capture / piracy**, an aggressive **pirate** stream beheads its neighbour, leaving an **elbow of capture**, a dry **wind gap**, and a **misfit** stream behind.
- Rejuvenation extras include paired **terraces**, **incised / entrenched** meanders, and a **knickpoint** that migrates upstream.

Stream	Follows
Consequent	Initial slope
Subsequent	Weaker beds / **faults** (trellis valleys)
Obsequent	Opposite to consequent (scarp face)
Resequent	Same direction as consequent, later
Insequent	No obvious control (dendritic on uniform rock)

N.6 Types of Deltas

A delta needs a large load, a quiet coast and a shallow shelf. Strong tides or waves favour an **estuary** instead.

Delta shapes — Teaching Card

- An **arcuate** (चापाकार) delta has a **bow / fan** front and many distributaries.
- The **Nile** (नील) builds an arcuate delta in **northern Egypt**.
- The **Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)** builds an arcuate delta in the **Sundarbans (सुंदरबन)**.
- The **Rhine** (राइन) builds an arcuate delta in the **Netherlands**.
- The **Niger** (नाइजर) builds an arcuate coastal delta in **Nigeria**.
- The **Hwang Ho** (ह्वांग हो) builds an arcuate delta on China's **Bohai** coast.
- A **bird's-foot** (पक्षी-पाद) delta sends long toe-like distributaries into quiet water.
- The **Mississippi** (मिसिसिपी) is the **only** bird's-foot example.
- A **cuspate** delta is **tooth / pointed** where waves from two sides pinch the mouth.
- The **Tiber** (टाइबर) in Italy is a cuspate example.
- The **Ebro** (एब्रो) in Spain is a cuspate example.
- An **estuarine** mouth fills a drowned funnel rather than a fan.
- The **Narmada** (नर्मदा) meets the Arabian Sea as an estuary.
- The **Tapi** (तापी) meets the Arabian Sea as an estuary.
- East-coast deltas sit on a wide shelf.

Moribund — two different questions

- A **moribund** (मृतप्राय) belt is the **dying** part of a delta, where old channels no longer carry the main flow.
- The **Bengal Delta** is split into three belts: **moribund**, **mature**, and **active**.
- If the stem says moribund delta is a **subdivision** of a named delta, the answer is the **Bengal Delta**.

- That moribund belt sits on the older **western** side of Bengal, after the main flow shifted east toward the **Padma**.
- If the stem asks which **river's** delta is the moribund or **abandoned lobe**, the answer is the **Cauvery** (कावेरी).
- The Cauvery delta is old and heavily irrigated, and it adds little new land at the sea.
- **Krishna (कृष्णा)–Godavari (गोदावरी)** is not the answer when the word is **subdivision**.
- **Krishna–Godavari** is not the answer when the word is **abandoned lobe**.
- **Mahanadi (महानदी)** is not the answer when the word is **subdivision**.
- **Mahanadi** is not the answer when the word is **abandoned lobe**.
- **Subdivision** means Bengal. **Abandoned lobe** means Cauvery. Each wording keeps its own answer.

Wording in the stem	What it asks	Answer
Subdivision of a named delta	Which delta is split into moribund–mature–active belts	**Bengal Delta**
Which **river's** delta is the moribund / abandoned **lobe**	Which delta is old and no longer growing	**Cauvery**
Either wording, with Krishna–Godavari or Mahanadi as the pick	Distractor on both stems	**Neither**

If the stem swaps the river — where the delta lies

The Mekong stem keeps a true source and moves the delta to the wrong country. The same test can use any river below.

River	Rises in	Delta lies in	Usual wrong place
Mekong (मेकांग)	Tibetan Plateau	**Southern Vietnam**	Southern Cambodia
Red River	Yunnan, China	**Northern Vietnam**	Southern Vietnam (that is the Mekong)
Irrawaddy (इरावदी)	Northern Myanmar	**Southern Myanmar**	Thailand
Chao Phraya	Northern Thailand	**Central Thailand**	Cambodia
Nile (नील)	White Nile: Lake Victoria; Blue Nile: Lake Tana	**Northern Egypt**	Sudan or Ethiopia
Indus (सिंधु)	Tibet, near Mansarovar	**Sindh, Pakistan**	Indian west coast
Ganga–Brahmaputra	Himalaya	**Bengal** (India and Bangladesh)	Assam alone
Godavari (गोदावरी)	Trimbakeshwar	**Andhra Pradesh**	Odisha or Tamil Nadu
Krishna (कृष्णा)	Mahabaleshwar	**Andhra Pradesh**	Karnataka at the sea
Mahanadi (महानदी)	Sihawa, Chhattisgarh	**Odisha**	Andhra Pradesh

River	Rises in	Delta lies in	Usual wrong place
Cauvery (कावेरी)	Talakaveri, Karnataka	**Tamil Nadu**	Karnataka at the sea
Narmada / Tapi	Amarkantak / Multai	**Estuary**, not a large delta	An arcuate fan
Danube (डेन्यूब)	Black Forest, Germany	**Black Sea**, Romania and Ukraine	Hungary or Austria
Rhine (राइन)	Swiss Alps	**Netherlands**	Germany only
Volga (वोल्गा)	Valdai Hills, Russia	**Caspian Sea**, Russia	Black Sea
Niger (नाइजर)	Guinea highlands	Inland delta in **Mali**; sea delta in **Nigeria**	Treating Mali as the sea mouth
Amazon	Andes	Broad **estuary in Brazil**	A fan delta in Peru
Hwang Ho	Tibetan Plateau	**Bohai**, China	A bird's-foot mouth
Mississippi	Lake Itasca, Minnesota	**Louisiana**, Gulf of Mexico	Any second bird's-foot river

- The **Mekong** rises on the **Tibetan Plateau**, and its delta lies in **southern Vietnam**.
- Cambodia stands on the Mekong, but the delta is not in southern Cambodia.
- The **Red River** delta lies in **northern Vietnam**.
- The **Irrawaddy** delta lies in **southern Myanmar**.
- The **Nile** delta lies in **northern Egypt**.
- The **Indus** delta lies in **Sindh, Pakistan**.
- The **Danube** delta meets the **Black Sea** in Romania and Ukraine.
- The **Rhine** delta lies in the **Netherlands**.
- The **Volga** delta lies on the **Caspian Sea**.
- The **Niger** has an inland delta in **Mali**.
- The **Niger** has a sea delta in **Nigeria**.
- The **Godavari** delta lies in **Andhra Pradesh**.
- The **Krishna** delta lies in **Andhra Pradesh**.
- The **Mahanadi** delta lies in **Odisha**.
- The **Cauvery** delta lies in **Tamil Nadu**.
- The **Amazon** meets the Atlantic in **Brazil** as a broad estuary.

Type	Shape / fact	Classic
Arcuate	Bow / fan	Nile, Ganga–Brahmaputra (Sundarbans), Rhine, Niger, Hwang Ho
Bird's-foot	Distributaries like toes	**Mississippi only**
Cuspate	Tooth / pointed	Tiber (Italy), Ebro (Spain)
Estuarine	Fills a drowned mouth	Narmada, Tapi

Type	Shape / fact	Classic
Moribund (subdivision wording)	Dying belt inside a three-belt scheme	**Bengal Delta**
Moribund (lobe wording)	Abandoned / ageing delta	**Cauvery**

Inline PYQ — UPPCS Prelims 2018, Q33

Bird's-foot delta is typically associated with:

A. Nile

B. Ganga–Brahmaputra

C. Mississippi

D. Cauvery

Show answer

Logic: Bird's-foot has one example. The Nile and the Ganga–Brahmaputra are arcuate. The Cauvery is an old delta, not this shape. **Ans: C** — The **Mississippi** is the only bird's-foot example.

Inline PYQ — UPPCS Prelims 2019, Q85

'Moribund Delta' is a subdivision of which of the following Delta?

A. Krishna–Godawari Delta

B. Mahanadi Delta

C. Bengal Delta

D. Cauvery Delta

Show answer

Logic: The stem says **subdivision**. That wording matches Bengal's moribund–mature–active scheme, not the separate Cauvery "ageing lobe" match-list. **Ans: C** — **Bengal Delta**. Cauvery is the usual wrong pick when students remember only the lobe tag.

Inline PYQ — UPPCS Prelims 2021, Q121

With reference to the Mekong, which of the following statements is/are correct?

1. It rises from the Tibetan Plateau.
2. Its delta lies in southern Cambodia.

A. 1 only

B. 2 only

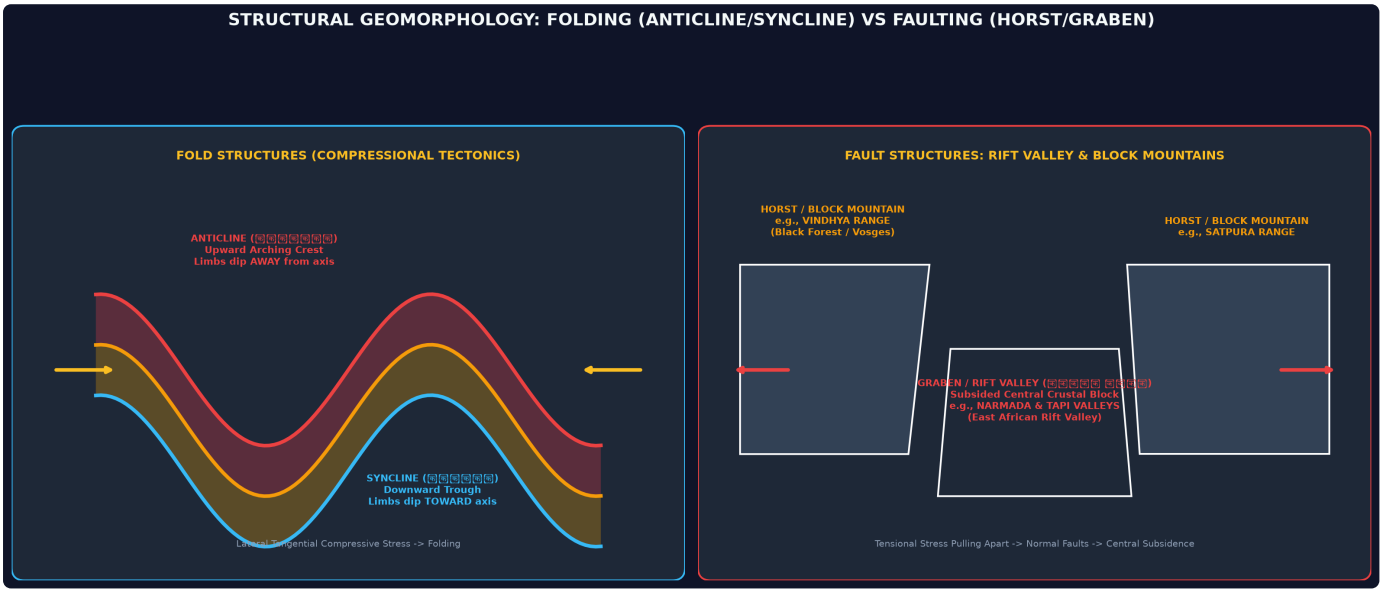
C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Logic: The rise on the Tibetan Plateau is correct. The delta country is the trap: southern Vietnam, not southern Cambodia. The same stem can name another river from the delta-country table above. **Ans: A** — Only statement 1. The Mekong delta lies in **southern Vietnam**.

N.7 Structural Basins & Fault Valley Drainage



A **structural basin** is a low enclosed by structure (syncline, rift or down-warp). Drainage often becomes **centripetal**.

Example	Type
Imphal basin	**Lacustrine** plain (old lake) — **not** loess / glacial / ordinary alluvial trap
Kashmir vale	Lacustrine / structural (Pir (पीर) Panjal (पीर पंजाल) wall)
Chhattisgarh / Michigan type	Structural/sedimentary basin

Depositional plains (trap set)

Plain	Agent
Alluvial	River (Indo-Gangetic)
Lacustrine	Old lake floor — **Imphal**, Kashmir
Glacial / outwash	Ice
Loess	Wind silt — **China**; not Imphal
Lava / volcanic	Flood basalt (Deccan surface)

- **Fault-valley drainage** follows subsequent streams along **faults**.
- Joints give **rectangular** patterns.
- Indian rifts include **Narmada, Tapi and Damodar**.
- The UPPCS fact “famous for fault valley drainage” is **Damodar**, not **Chambal** (Chambal is ravine/gully country).
- World rift pairs include the Rhine, Dead Sea/Jordan and East Africa.

Inline PYQ — UPPCS Prelims 2019, Q8

Imphal basin is an example of:

A. Loess plain

B. Lacustrine plain

C. Glacial plain

D. Only aeolian erg

Show answer

Ans: B — Old lake / lacustrine floor.

Inline PYQ — UPPCS Prelims 2019, Q9

Which of the following rivers is famous for fault valley drainage?

A. Chambal

B. Damodar

C. Betwa (बेतवा)

D. Ken

Show answer

Ans: B — **Damodar**. Chambal = ravines.

Structural landforms — Teaching Card

Structural landforms are hills and edges shaped by **tilted beds**, **hard caps**, or **fault blocks** — not by a river carving alone.

Cuesta vs hogback (क्वेस्टा) (same family — different dip)

- A **cuesta** forms where hard and soft rock layers are stacked like a sandwich and dip gently.
- One side is a **gentle dip slope** (you walk with the tilt of the beds).
- The other side is a **steep scarp** where soft rock under the hard cap has been worn back.
- A **hogback** is the same idea when the beds dip **steeply**.
- On a hogback, **both sides look steep** — there is no gentle walk-up slope.

Form	Dip of beds	Side view
Cuesta	Gentle	One gentle slope + one steep scarp
Hogback	Steep	Steep on both sides

Mesa vs butte (मेसा) (same family — different size)

- A **mesa** is a broad, flat-topped remnant protected by a **hard cap rock**.
- Soft rock around it erodes away; the hard top keeps a table-like plateau.
- A **butte** is the **smaller leftover** of a mesa after more sides are cut back.
- Rule of thumb: **mesa = broad table**; **butte = narrow leftover**.

Form	Size feel	Cap
Mesa	Broad flat top	Hard cap still wide
Butte	Smaller / pillar-like	Same idea, much less area

Fault-block & edge forms

- A **horst** is an **uplifted block** between faults — higher ground.
- A **graben** is a **down-dropped block** (rift floor) between faults — lower ground.
- An **escarpment** is a long **steep edge** — often the rim of a plateau or a fault scarp.
- Keep the pair: horst = up; graben = down (like a dropped floor between walls).

Drainage patterns — Teaching Card

A **drainage pattern** is the map shape of rivers. The shape tells you what **controls** the ground — rock type, folds, joints, or slope.

Tree and grid patterns

- **Dendritic** (वृक्षाकार) looks like a **tree** with many irregular branches.
- It develops on **homogeneous rock** (same resistance everywhere) — no strong fold or joint grid to force streams into lines.
- The **Ganga (गंगा)** on the northern plain follows a dendritic pattern.
- The **Indus (सिंधु)** on the plain follows a dendritic pattern.
- The **Godavari (गोदावरी)** follows a dendritic pattern.
- The **Mahanadi (महानदी)** follows a dendritic pattern.
- The **Krishna (कृष्णा)** follows a dendritic pattern.
- The **Kaveri (कावेरी)** follows a dendritic pattern.
- **Trellis** (जालनुमा) looks like a **garden trellis / lattice** — main streams in long parallel valleys, short cross-links.
- It develops on **folded** belts of **hard and soft rock** alternating — soft belts become strike valleys; hard belts become ridges.
- The Indian trellis ground is the old folded belt of **Singhbhum (सिंहभूम)** on the **Chotanagpur (छोटानागपुर)** plateau.

- Memory: **same rock → tree (dendritic)**; **folds → lattice (trellis)**.

From a high centre or into a bowl

- **Radial** (अरीय) streams run **outward** from a high centre — like spokes from a hub.
- Typical controls: **dome**, **volcano**, or a highland such as **Amarkantak (अमरकंटक)**.
- The **Narmada (नर्मदा)** leaves Amarkantak toward the **west**.
- The **Son (सोन)** leaves Amarkantak toward the **Ganga**.
- The Mahanadi rises at **Sihawa**, so it is not an Amarkantak radial river.
- **Girnar (गिरनार)** hill in Gujarat also sends streams outward.
- The **Mahaweli (महावेली)** drains **Sri Lanka**, where streams radiate from the central highlands.
- **Centripetal** (केन्द्राभिमुख) is the opposite — streams run **inward** into a **closed basin**.
- Streams of the **Imphal** basin drain inward toward **Loktak (लोकतक)**.
- The **Mendha (मेंढा)** drains inward toward **Sambhar (सांभर)** lake.
- The **Rupangarh (रुपनगढ़)** drains inward toward Sambhar lake.

Joints, slope, rings, and chaos

- **Rectangular** (आयताकार) drainage bends at near-right angles because streams follow **joints** and **faults**.
- Streams on the jointed **Vindhyan (विंध्य)** rocks follow a rectangular pattern.
- **Parallel** (समानांतर) drainage has many streams running side by side on a **steep, uniform slope**.
- The **Sharavati (शरावती)** follows a parallel pattern down the **Western Ghats (पश्चिमी घाट)**.
- **Annular** (वलयाकार) drainage forms **ring-like** courses where soft beds sit in **concentric** arcs around a dome or basin.
- The usual annular example is the **Black Hills** of South Dakota. No single Indian river carries this tag.
- **Deranged** (अव्यवस्थित) drainage is messy and poorly organised on a **new glacial dump** (kame-and-kettle / recently ice-left ground) before a clear network develops.
- Deranged ground, such as the **Canadian Shield**, has no single master river.

Pattern	Looks like	Control (why)	River / ground
Dendritic	Tree branches	Homogeneous rock	**Ganga** and **Indus** plains; **Godavari**; **Mahanadi**; **Krishna**; **Kaveri**
Trellis	Lattice / garden trellis	Folded hard–soft belts	**Singhbhum** (Chotanagpur)
Radial	Spokes outward	Dome / volcano / highland	**Narmada** west and **Son** toward the Ganga from **Amarkantak**; Girnar; **Mahaweli** (Sri Lanka)
Centripetal	Spokes inward	Closed basin	**Loktak** (Imphal basin); **Mendha** and **Rupangarh** into **Sambhar**
Rectangular	Right-angle bends	Joints / faults	Jointed **Vindhyan** streams
Parallel	Side-by-side lines	Steep uniform slope	**Sharavati** (Western Ghats)

Pattern	Looks like	Control (why)	River / ground
Annular	Rings	Concentric weak beds	**Black Hills** — no single Indian river
Deranged	Chaotic / unfinished	New glacial dump	**Canadian Shield** — no master river

Inline PYQ — UPPCS Prelims 2024, Q31

Trellis drainage pattern typically develops on:

A. Homogeneous crystalline rock only

B. Folded hard–soft rock belts

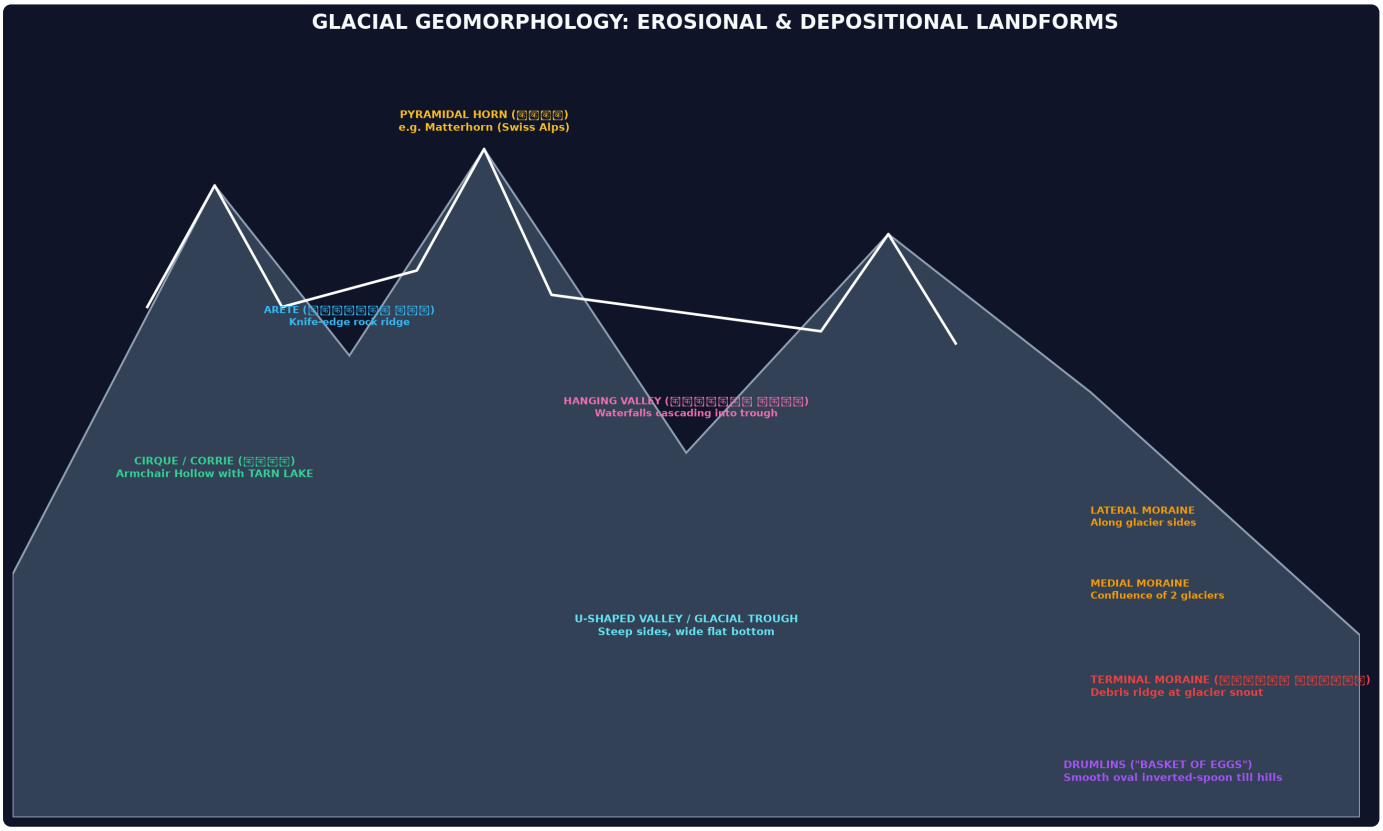
C. Only closed desert basins

D. Only ice sheets

Show answer

Ans: B — Folded structure → **trellis**; homogeneous rock → dendritic.

N.8 Glaciers & Glacial Landforms



A glacier is a mass of ice that moves under its own weight. Fact the **U-valley vs V-valley** pair, the **fjord vs ria** pair, and the depositional till suite.

Glacier types — Teaching Card

- A **cirque glacier** sits in an armchair hollow at the head of a mountain valley — ice starts here.
- A **valley / alpine glacier** is a tongue of ice flowing down a mountain trough.
- A **piedmont glacier** forms when valley glaciers leave the mountains and **coalesce** as a broad ice sheet on a plain.
- An **ice sheet / continental glacier** covers a whole region — **Greenland** and **Antarctica** are the living giants.
- An **ice cap** is a smaller dome of ice (Iceland type) — not continent-wide.
- An **ice shelf** is the floating fringe of ice extending from land onto the sea.

Type	Quick recall
Cirque glacier	Armchair hollow
Valley / alpine	Tongue in a trough
Piedmont	Coalesced on a plain
Ice sheet	Greenland, Antarctica
Ice cap	Smaller dome
Ice shelf	Floating fringe

- Accumulation sits above the **snowline**; ablation sits below it.
- **Firn / névé** is compacted snow on the way to glacier ice.
- A **bergschrund** cracks open at the headwall; crevasses and seracs mark the tongue.
- A **nunatak** is a peak sticking through ice.

Erosional glacial landforms — Teaching Card

- A **cirque** (corrie / cwm) is the armchair hollow scooped at a glacier's head; a lake left in it is a **tarn**.
- An **arête** is the knife-edge ridge left between two neighbouring cirques.
- A **horn** is a pyramidal peak where three or more arêtes meet — the **Matterhorn** is the classic picture.
- A **U-shaped** trough is carved by a valley glacier; a river cuts a **V-shaped** valley instead.
- A **hanging valley** sits higher than the main trough because the tributary glacier was smaller — its lip often holds a waterfall after the ice melts.
- A **fjord** is a glacial trough drowned by the sea (Norway, Chile, New Zealand).
- A **roche moutonnée** is bedrock smoothed and abraded on the up-ice side and plucked on the lee side.
- A **crag-and-tail** pairs a hard rock knob with a debris tail stretching down-ice.

Form	Quick recall
Cirque / tarn	Armchair hollow; lake inside
Arête	Knife ridge between cirques

Form	Quick recall
Horn	Pyramidal peak (Matterhorn type)
U-trough	Glacier (≠ river V)
Hanging valley	Higher tributary floor → waterfall
Fjord	Drowned glacial trough
Roche moutonnée	Abraded + plucked bedrock
Crag-and-tail	Hard rock + lee debris

Depositional glacial landforms — Teaching Card

- A **moraine** is glacial debris dumped by ice: **lateral** along the sides, **medial** where two glaciers join, **terminal** at the snout, plus **ground** and **recessional** belts.
- A **drumlin** is a streamlined mound of till; the blunt end faces **up-ice** and the taper points down-ice.
- An **esker** is a long, sinuous ridge of sand and gravel left by a meltwater stream under or within the ice.
- A **kame** is a mound of ice-contact sand and gravel dumped against or in melting ice.
- An **outwash plain** (sandur) is the meltwater apron of sorted sediment beyond the terminal moraine.
- An **erratic** is a far-travelled boulder dropped far from its source rock.
- A **kettle** forms when a buried ice block melts and leaves a pit that may hold a lake.

Form	Quick recall
Moraine	Lateral, medial, terminal, ground
Drumlin	Streamlined till; blunt up-ice
Esker	Sinuous subglacial sand–gravel
Kame	Ice-contact mound
Outwash / sandur	Meltwater plain beyond snout
Erratic	Far-travelled boulder
Kettle	Melt pit → lake

Himalayan glaciers act as regional water towers for the Indo-Gangetic rivers.

N.9 Aeolian Landforms

Wind needs dry ground, sparse vegetation and a sand or silt supply — hot deserts and cold loess belts both qualify.

Wind work and erosional forms — Teaching Card

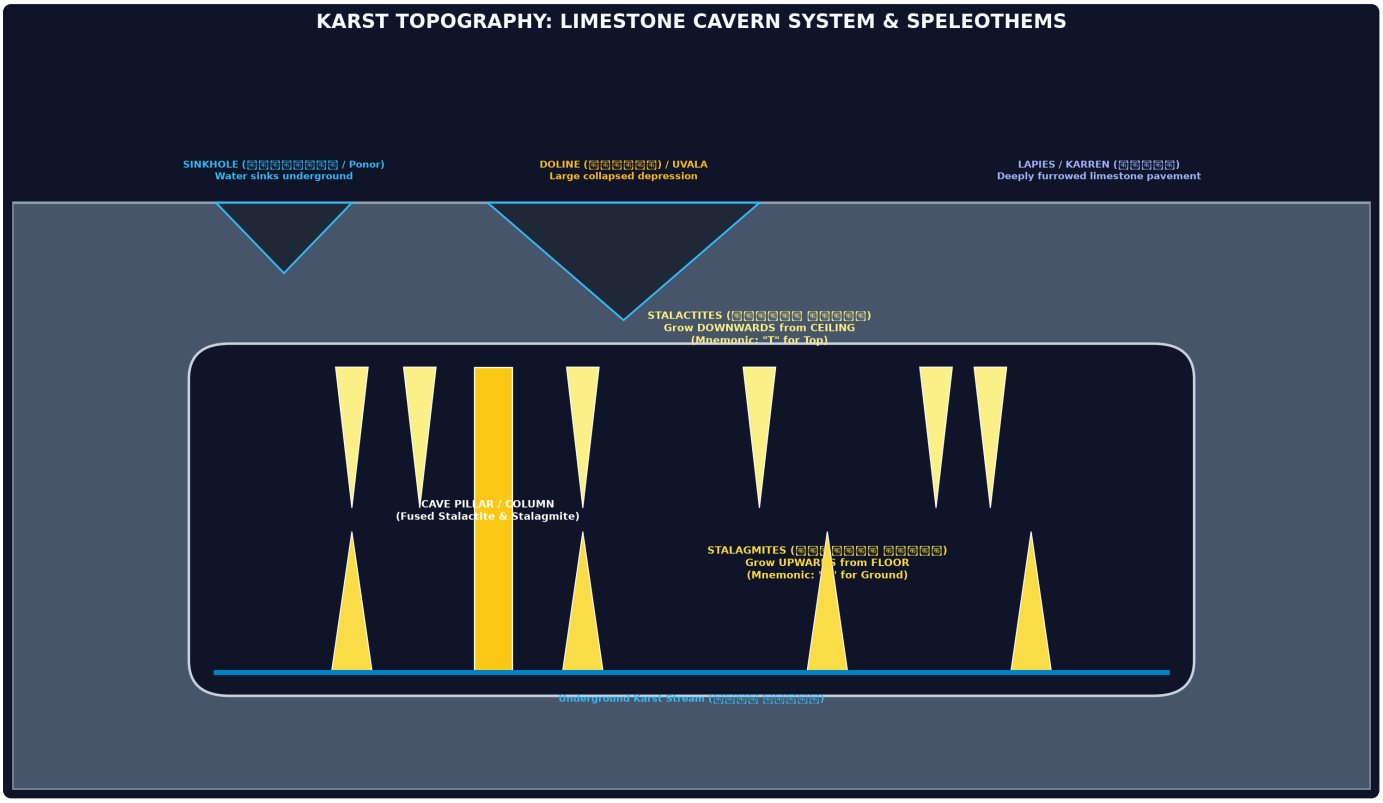
- Wind moves sand mainly by **surface creep** and **saltation** (the hopping bounce); fine dust travels in **suspension** and builds **loess**.
- **Deflation** lifts fine dust and lowers desert floors into hollows and **playas**.
- A **yardang** is a streamlined ridge carved **parallel to the wind** by abrasion — elongated and wind-aligned.
- A **zeugen** is a tabular residual rock sitting (बैठक) on a hard cap after softer rock around it is stripped.
- **Mushroom / pedestal** rocks stand on a narrow stem where near-ground abrasion undercuts the base.
- **Ventifacts** are pebbles polished and faceted by sandblast.
- An **inselberg** is an isolated residual hill rising from a worn plain; **pediments** are gently sloping rock aprons at mountain feet.
- Desert pavement (**reg**) is a stone armour left after fines are blown away; bare-rock desert is **hamada**.
- An **erg** is a sand sea — a depositional province of dunes, not an erosional hollow.

Dunes and loess — Teaching Card

- A **barchan** is a crescent dune with horns that point **downwind** (with the wind).
- A **parabolic** dune is also crescent-shaped, but its horns point **upwind** (into the wind) — often coastal and vegetation-anchored; it is the **opposite** of a barchan.
- A **seif / longitudinal** dune is a long ridge running **parallel** to the dominant wind.
- A **transverse** dune has its crest **across** the wind where sand supply is plentiful.
- A **star / pyramidal** dune grows under **multi-directional** winds into a peaked hub with radiating arms.
- **Loess** is wind-laid silt — fertile and thick on China's loess plateau.
- The **Imphal** basin is **lacustrine** (old lake floor), **not** loess — do not swap the two plains.

Form	Quick recall
Barchan	Crescent; horns **downwind**
Parabolic	Crescent; horns **upwind** (opposite barchan)
Seif / longitudinal	Ridge parallel to wind
Transverse	Crest across wind
Star / pyramidal	Multi-directional winds
Loess	Wind silt; China — **not** Imphal
Erg	Sand sea

N.10 Coastal Landforms (+ karst)



Waves run **swash** up the beach and **backwash** down. **Longshore drift** moves sediment along the shore and builds spits.

- The **erosional** coastal chain runs cliff, then wave-cut platform, then notch, then cave, then **arch**, then **stack**, then stump.

Depositional coastal forms — Teaching Card

Longshore drift and wave deposition build soft coastal forms in shallow water.

- A **beach** is a wave-built strip of sand or shingle along the shore.
- A **spit** is a narrow ridge of sand or shingle attached at **one** end to the coast and free at the other — it grows in the direction of longshore drift.
- A **bar** is a ridge that closes or nearly closes a bay mouth and often traps a quiet **lagoon** behind it.
- A **tombolo** is a bar that ties an **island** to the mainland (or joins two islands).
- A **hook** is a spit tip bent into a curve by wave refraction around the free end.
- A **barrier island** is an elongated offshore island built roughly parallel to the mainland coast.

Form	Quick recall
Beach	Wave-built sand / shingle strip
Spit	Attached at one end; free tip
Bar	Closes a bay; lagoon behind
Tombolo	Island tied to mainland

Form	Quick recall
Hook	Curved spit tip
Barrier island	Offshore, parallel to coast

Coral reefs — Teaching Card

Coral reefs grow in clear, warm, shallow seawater. Muddy delta mouths are poor reef sites.

Three reef types (Darwin sequence)

- A **fringing** reef grows **against the shore** of an island or coast — no wide lagoon yet.
- A **barrier** reef stands **offshore**, with a lagoon between the reef and the land.
- An **atoll** is a coral reef that encloses a **central lagoon**, with **no large volcanic island left in the middle**.
- Darwin's **subsidence** idea puts them in order: **fringing** → **barrier** → **atoll** as the volcanic island sinks and coral keeps growing upward.

Shape of an atoll (UKPCS fact)

- The classic **shape of an atoll** is **horseshoe-shaped** or **ring-shaped**.
- The reef rim runs around a shallow **lagoon** in the middle — like a ring, or like a horseshoe if one side is open or broken by channels.
- It is **not** rectangular, rod-shaped, or triangular as the form name.
- Real atolls can look oval or slightly irregular, but the form answer stays **horseshoe / ring**.
- India's **Lakshadweep** (लक्षद्वीप) islands are classic **coral atolls**; Maldives atolls follow the same ring-lagoon idea.

Reef type	Where it sits	Lagoon?
Fringing	Against the shore	Little / none
Barrier	Offshore, parallel to land	Yes — between reef and land
Atoll	Ring / horseshoe in open sea	Yes — **central** lagoon; island mostly gone
Coast type	Meaning	
---	---	
Fjord	Drowned **glacial** trough	
Ria	Drowned **river** valley (non-glacial)	
Dalmatian	Concordant drowned ranges // coast (Adriatic type)	
Emergent	Raised beach / wave-cut platform now inland	
Submergent	Ria / fjord / drowned mouth	

Karst needs limestone plus carbonation.

- Forms include swallow hole, **doline / sinkhole**, uvala, **polje**, cave, limestone pavement (clint / gryke), disappearing stream and resurgence.
- A **stalactite** hangs from the **roof**.
- A **stalagmite** grows from the **floor**.
- A pillar forms when they meet.
- Classic world belt is the Dinaric Karst.
- India pairs include **Meghalaya** (मेघालय) caves, Borra (AP), Belum (AP), **Parner Taluka** (Ahmadnagar district, Maharashtra), and **Bastar District** (Kanger Ghati caves, Chhattisgarh).
